

Physiological Response to Phantom Touch in VR: Investigation of the Impact on Electrodermal Activity and Heart Rate

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Figure 1. Participant looking into the virtual mirror while the researcher is touching their arm to develop Phantom Touch

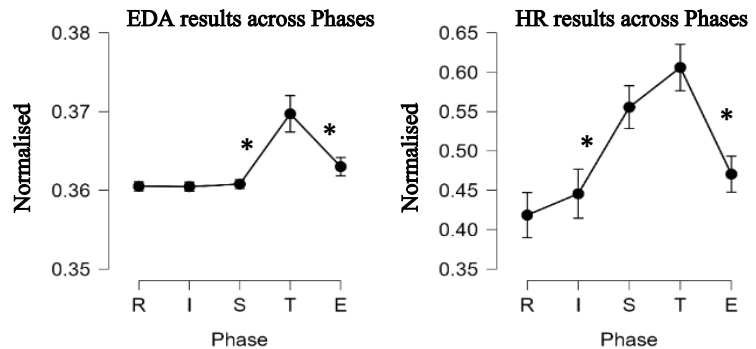


Figure 2. EDA and HR response among the participants showing significant increase from phases S – T and T – E in EDA, and in I – S and T – E in HR

ABSTRACT

Phantom Touch in VR - a phenomenon which originated in the Metaverse which allows users to experience social touch in VR without any stimuli or haptic devices. Previous research has primarily relied on participants' subjective responses to determine whether they feel Phantom Touch. However, objective physiological response to the phenomenon is scarce. This work presents findings on electrodermal activity and heart rate during Phantom Touch in VR development experiments, alongside subjective response to confirm the physiological results. The findings reveal measurable physiological changes associated with Phantom Touch, indicating heightened arousal in response to the newly developed sensations.

Index terms: Phantom Touch in VR, social touch, pseudo-haptic touch, physiological response, Metaverse, VRChat

1 INTRODUCTION

The Phantom Touch in VR phenomenon, originating on the Metaverse platform VRChat, enables users to experience touch sensations without the use of haptic devices or physical stimuli in real life [1]. Previous research has primarily focused on the subjective experiences of individuals who report feeling Phantom Touch, with several studies relying on questionnaire-based measures to understand the phenomenon [2]. Studies that aimed to

induce Phantom Touch in experimental settings have similarly focused on subjective reports, often evaluating whether participants experience the sensations at all [1, 3]. The pioneering research that first investigated how Phantom Touch could be developed introduced a method that enabled participants to generate touch sensations within a controlled experimental setting [1]. The study relied primarily on subjective measures, asking participants whether they felt the sensations on the body parts being touched in VR. For participants who reported developing Phantom Touch, heart rate was recorded before and during the interaction in VR. The findings suggested that Phantom Touch interaction led to a decrease in heart rate. However, the study had a small sample size ($N = 10$), which may have affected the reliability and generalizability of its conclusions.

A subsequent study further explored Phantom Touch by examining the subjective reports of participants who touched their own hand with a virtual stick in VR, focusing on qualitative descriptions of the sensations [3].

However, few studies have examined the physiological response associated with Phantom Touch development in an experimental setting. Building on these prior works, this research advances experimental research on Phantom Touch in VR by investigating the physiological responses associated with the development of the phenomenon. Specifically, the measures of Electrodermal Activity (EDA) and Heart Rate (HR) were observed, as these are common methods of investigating the autonomic nervous system (ANS) response to stimuli [4].

The main contribution of this research is a systematic demonstration that Phantom Touch alters the sympathetic nervous system (SNS) during interaction in VR, indicating an increase in arousal. These results provide new insight into the physiological response to the phenomenon, demonstrating that Phantom Touch extends beyond a purely subjective perceptual experience.

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2 EXPERIMENT PROCEDURE

The Head-to-Toe (HTT) method for Phantom Touch development, in which sensations are induced by sequentially touching each body part, starting from the head and progressing down the body, was adopted from the work of Alexdottir et al., [1] as it has been shown to be effective in eliciting Phantom Touch sensations. A calm virtual environment was selected for the experiment to support relaxation and allow participants to focus on the process of Phantom Touch development. A virtual mirror was also enabled in the virtual world (as shown in Figure 1), as prior research has demonstrated its effectiveness in facilitating sensation development [1]. Participants were allowed to freely choose their own avatar from the VRChat avatar library, as previous findings suggest that avatar autonomy and relatability can promote Phantom Touch development [1]. In addition, participants selected an avatar for the touch initiator from four commonly used Metaverse avatar types, ensuring comfort during the touch interactions. An Oculus Quest 2 was worn by the researcher and participant, and the physiological responses were recorded using an EMOTIBIT device which was attached to the participant's forearm, measuring EDA in microsiemens (μS) [5] and HR in beats per minute [6]. During the experiment, the participant and the touch initiator (the researcher) were in separate physical rooms to prevent audio feedback and to enhance participant comfort, immersion, and focus during the experiment. The experiment had five consecutive sessions to allow participants to develop Phantom Touch, as previous research has shown that several sessions are required [1].

3 RESULTS

30 participants took part in the study, with 28 used for analysis, as 2 did not develop any Phantom Touch sensations (Female = 15, Male = 13) (Age 16-20 = 10, 21-25 = 9, 26-30 = 5, 31-35 = 3, 36-40 = 1). All participants had no prior experience with Phantom Touch or VR.

To determine whether fluctuations in the signal could indicate the effect of Phantom Touch, an average was calculated from a representative phase within each section. Analysis was split into five distinct phases: Rest outside of VR prior to putting the headset on (R), VR idle time - when the participant enters VR (I), start of the experiment (S), Phantom Touch interaction (T), and end of the experiment (E). For phase T, only one instance of Phantom Touch interaction was analysed rather than every single touch point from the HTT method. Furthermore, every phase used for analysis was taken from timeframes where the participant was not moving or transitioning from one section to another to reduce movement noise and bias in the data (e.g. standing up to put the headset on from phases R to I).

All data were normalised using the min-max scaling method across all sessions for a given participant so that data would be on a comparable scale between participants [7].

The data were analysed using repeated-measures ANOVA with the factors Session (1 to 5) and the five Phases. Greenhouse-Geisser correction was applied whenever the sphericity assumption was violated. In EDA, there was a significant main effect of Session, $F(3.3, 59.1) = 2.858, p = 0.041, \eta^2 = 0.137$. Post hoc tests showed a significant change between sessions one ($M = 0.62, SE = 0.099$) and two ($M = 0.308, SE = 0.077$) ($t = 2.63, p = 0.017$), after which the sessions showed no significant changes (ranged from 0.247 to 0.381) (all $t_s < 0.695, p_s > 0.287$). The main effect of Phase was also significant, $F(1.3, 23) = 10.069, p = 0.002, \eta^2 = 9.148 \times 10^{-5}$. As seen in Figure 2, Post hoc comparisons demonstrated a significant increase from S ($M = 0.361, SE = 0.019$) to T ($M = 0.37, SE = 0.019$) ($t = -3.511, p = 0.002$), then a significant drop from T to E ($M = 0.363 \pm 0.019$). There was no difference in the R-I-S period (ranged from $M = 0.36$ to 0.361 ± 0.019) ($t_s < 0.221, p_s > 0.218$).

The interaction between Sessions and Phase was not significant, $F(3.6, 64.2) = 0.638, p = 0.619, \eta^2 = 2.617 \times 10^{-5}$.

Analysis of the HR data revealed significant differences between the Phases, while no statistically significant changes were observed across sessions. There was a significant main effect of Phase, $F(2.6, 43.4) = 7.894, p < 0.001, \eta^2 = 0.08$. As seen in Figure 2, the Post hoc test identified a significant increase between phases I ($M = 0.446, SE = 0.04$) and S (0.556 ± 0.4) ($t = -2.613, p = 0.018$), and a significant decrease between T (0.606 ± 0.039) and E (0.471 ± 0.027), with no significant change between R and I (ranged from 0.418 to 0.446 ± 0.036 to 0.04) ($t = -0.823, p = 0.422$) or S and T (ranged from 0.0556 to 0.606 ± 0.39 to 0.4) ($t = -1.871, p = 0.079$). No significant differences were observed between Sessions, $F(2.8, 46.6) = 2.584, p = 0.069, \eta^2 = 0.055$, nor in the interaction between Session and Phase, $F(7.3, 123.5) = 1.038, p = 0.409, \eta^2 = 0.019$.

4 CONCLUSION

The results indicate an increase in SNS activity during the Phantom Touch interaction phase (T), showing heightened arousal compared to the phases in which participants were not touched (R, I, S, E). Additionally, the results showed no significant differences between sessions in HR, appearing only in EDA between the first two sessions, suggesting participants acclimated to the sensations after the second session. This indicates that the intensity of the sensations did not greatly fluctuate with repeated exposure. These findings offer new insight into the physiological response to Phantom Touch, demonstrating that the phenomenon produces measurable autonomic changes beyond subjective reports. However, these results need to be confirmed through a control group experiment where participants do not experience Phantom Touch in VR, in order to determine whether the observed physiological responses can be attributed specifically to the occurrence of the phenomenon.

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